Improving the Metabolic Efficiency of the Gait Cycle in the NASA Space Suit Through Biomechanical Interventions

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Overview

The extravehicular mobility unit (EMU) causes extensive mobility and fatigue issues in astronauts, causing decreased efficiency on planetary missions. In this project, we explore one method of passive force application in order to lessen metabolic cost during the gait cycle. Using advances in the realm of biomechanics, the passive device has been designed to seamlessly integrate in the EMU and provide metabolic savings during the gait cycle.

Background

Current Challenges

- Gas-pressurization in space suits leads to resistive joint torques, overall fatigue, and injuries
  → Total work to fatigue time decreases by 50% [1]
- Total resistive torque at knee joint is ~ 10 Nm [2]
- The walking gait cycle is thus impaired by the resistivity

Solution Inspiration

- Natural stabilizing factors like the swinging of arms and harmonic oscillation in springs
- Predicated on terrestrial ideas like swimming technical suits, portable exotendons [3], and exoskeletons [4]; as well as EMU based solutions like in Diaz et al [2]

Design Challenges

- NASA is not likely to change suit architecture to accommodate a bulkier robotic exoskeleton
- Robotic exoskeletons require large power budgets

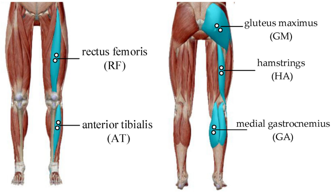

Figure 1: Diagram Highlighting the main muscle groups involved in the walking cycle
Taken from: Y. Liu et al. Applied Sciences. 2020; 10(14):4890

Design Requirements

- Must alter the existing suit as little as possible → be able to integrate into the liquid cooling and ventilation garment (LCVG)
- Must fit within the current envelope of the EMU → Restricts mechanism to less than an inch in thickness
- Requiring electrical power is too large of a tradeoff for not enough reward → Passive system
- Materials must be relatively simple → We do not have access to highly specialized textile manufacturing
- Must be able to overcome at least half of the resistive torque ~5 Nm

Prototype

Figures 2-6: Force application prototype using neoprene wetsuit to mimic compressive fit and thickness of LCVG garment. Use of easily interchangeable elastic bands to vary testing weights.

Testing Strategy

1. Measured and calculated the K constants for various samples of elastic materials to have a reference point and compare to our ideal and desired value.
2. Developed a foot and knee harness to test various elastic bands. The elastic bands were attached at the hip ,using a belt, and the other end was attached to a force meter held by the harness.
3. Captured the motion with a camera and recorded readings of the force meter with time intervals of 0.10 seconds.

Results

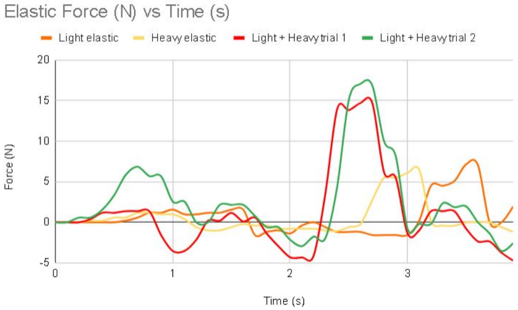

Figure 7: Measured force exertion over time for elastics with varying resistance levels.

Conclusion

Our results show that the basic elastic band prototype is able to exert up to 8 Nm of torque. This demonstrates that an elastomer attached to a suit successfully exerts a force that can be utilized in EMU contexts. In future iterations, we will be further testing band placement as well as more cohesive integration into a suit.

Acknowledgements

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References: [1] L. Gonzalez et al. Ergonomics. 2002;45:7, 484-500. [2] A. Diaz-Artiles et al. "SmartSuit" NIAC. 2020. [3] L. Cheng et al. R. Soc. Open Sci. 2021; 8:211266. [4] T. Zhou et al. IEEE Transaction on Neural Systems and Rehabilitation Engineering. 2021; 29:662-672.